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Resistive switching behaviors of oxygen-rich TaO_x films prepared by reactive magnetron sputtering

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ABSTRACT

In this work, Ta_2O_5 films were first deposited on Si substrates by reactive magnetron sputtering of a Ta metal target at various substrate temperatures, RF powers and sputtering pressures. The crystal characteristics of these films can be effectively tailored by controlling the sputtering process. Based on the optimized process parameters, tantalum oxide (TaO_x) films with different oxygen component content were sputtered on ITO buffered Si substrates and comparatively investigated. The results show that the film with Ta/TaO_x/ITO structure has a resistance switching (RS) behavior and its conduction mechanism is closely related to the O^{2-}/O concentration related to the oxygen partial pressure at the dielectric layer and electrode interface. This study provides an in-depth understanding of the component/structure design and structure-activity relationship for high-performance TaO_x-based resistive memory.

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KEYWORDS
TaO_x films; different oxygen concentration; magnetron sputtering; bipolar switching; RS mechanism

1. Introduction

As one of the four fundamental circuit elements, the memristor has attracted widespread attention and discussion due to its simple sandwich structure, nanoscale size, storage capacity and variable resistance [1,2]. Resistance random access memory (RRAM) has emerged as an attractive memory technology with high application prospects, thanks to its advantages of fast speed, high density, nonvolatile, multi-value storage, and multi-dimensional storage [3–5]. As a high-density random access memory embedded in various systems, this device plays a critical role in facilitating communication between edge devices and data centers [6,7]. A variety of thin-film materials exhibiting excellent switching characteristics have been used for fabricating these memory devices [8–10].

Recently, most research on the RS mechanism in oxide films has combined the space charge limited current (SCLC) mechanism with the presence of crystal defects (oxygen vacancies) that naturally exist in oxide films [11–13]. Several models have been constructed to attempt to explain the conduction mechanism in TaO_x films [14–17]. Consequently, investigation of the oxygen concentration in the oxide dielectric layer remains a valuable and open issue. In recent years, a variety of TaO_x-based memristors with different structures, including different TaO_x films, electrode materials and insertion layers, have been designed and investigated to explain the reversible RS phenomenon. The most fundamental factor leading to reversible RS in TaO_x films is the presence of oxygen non-stoichiometry and oxygen-deficient oxide phases [18,19]. However, most studies in the investigation of the RS mode polarity of TaO_x films prepared using Ta_2O_5 target, and less effort has been made in the study on oxygen concentration in TaO_x films prepared using Ta target [20–23].

In this study, we report the process optimization of preparing high-quality TaO_x films via radio frequency reactive magnetron sputtering using Ta target, focusing on the preparation of TaO_x films with different oxygen concentrations. Subsequently we investigate the bipolar memory switching behavior of Ta/TaO_x/ITO sandwich structures under different oxygen concentrations. Finally, we discuss the RS mechanism by considering the migration of metal ions in solid electrolytes and electrochemical reactions on electrodes.

2. Experimental section

TaO_x films were deposited on (100) Si substrates using a radio frequency reactive magnetron sputtering (RF-RMS) from a Ta metal target (99.95% in purity, 50 mm in diameter). The deposition process was performed in a mixed O_2 and Ar atmosphere with different content of O_2 (10%, 15%, 20%, 25% and 30%). Before being loaded into the chamber, the Si substrates ($1 \times 1 \times 0.05\ cm^3$) were cleaned successively in acetone,

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absolute ethanol and deionized water with an ultrasonic cleaner. For RS devices, an 80-nm-thick ITO layer was also sputtered from an ITO ceramic target to serve as the bottom electrode (BE).

The microstructure and composition of the TaO_x films were characterized by using X-ray diffractometry (XRD, D8 Advance A25), energy dispersive spectrometer (EDS), and scanning electron microscope (SEM, SU8000). The room temperature switching properties were measured by using a semiconductor parameter analyzer (Keithley +4200A-SCS). Prior to the electrical measurements, Ta top electrodes (TE) with a diameter of 200 μm (using a shadow mask) were deposited onto the surface of our TaO_x films via RF-RMS.

3. Results and discussion

In order to investigate the influence of the sputtering process and optimize the crystalline microstructure of TaO_x films, TaO_x films were fabricated under a variety of deposition conditions. The XRD patterns for the TaO_x films were displayed in Figures 1(a,c). It was clearly observed that Ta_2O_5 exhibits three main diffraction peaks corresponding to the (110), (200), and (086) crystal planes, with 2θ diffraction angles of 27.2°, 28.3°, and 56.3°, respectively [24,25]. In addition, some other peaks of Ta_2O_5, such as (111) also can be found in some XRD patterns. It indicated that the crystal characteristics of Ta_2O_5 films can be tailored effectively by controlling the sputtering process parameters (deposition temperature, sputtering power and gas pressure).The XRD patterns of TaO_x films deposited at various temperatures are shown in Figure 1(a), Where the RF power and gas pressure remained constant at 100 W and 1.0 Pa, respectively. XRD analysis revealed that when the deposition temperature reached 600°C or higher, crystalline phases were detected in Ta_2O_5 films at room temperature. This suggests that the crystallization temperature of Ta_2O_5 film must exceed 600°C. However, it is important to note that the crystallization process may also result in the formation of impurity-phases, such as incompletely oxidized Ta_5Si_3. At relatively low crystallization temperatures (600–700°C), the migration rate of atoms/clusters sputtered onto the substrate surface is relatively slow, which is not conducive to the growth of nucleation grains. As a result, the film displays poor crystallization performance and a relatively high concentration of defects. During the migration and nucleation process, silicon atoms are more likely to diffuse into the film, forming Ta_5Si_3 impurity phases. As the temperature continues to rise, the oxidation reaction in the reactive sputtering process becomes more sufficient, the crystallinity of Ta_2O_5 film is improved and the impurity-phase disappear. This may be mainly due to the decrease in defect density and inhibition of interface diffusion. Therefore, it can be concluded that the temperature has a significant influence on the growth process of Ta_2O_5 thin films. When the temperature is low (below 600°C), the film cannot nucleate and crystallize. At relatively low crystallization temperature (600–700°C), impurity-phases are generated in the film. When the temperature is above 700°C, a pure-phase Ta_2O_5 film can be obtained, but the temperature should not be too high, as excessively high temperature may reduce crystalline quality of the pure-phase film to some extent.

The deposition power will directly affects the bombardment energy of sputtered species, thus affecting their nucleation and growth processes after reaching

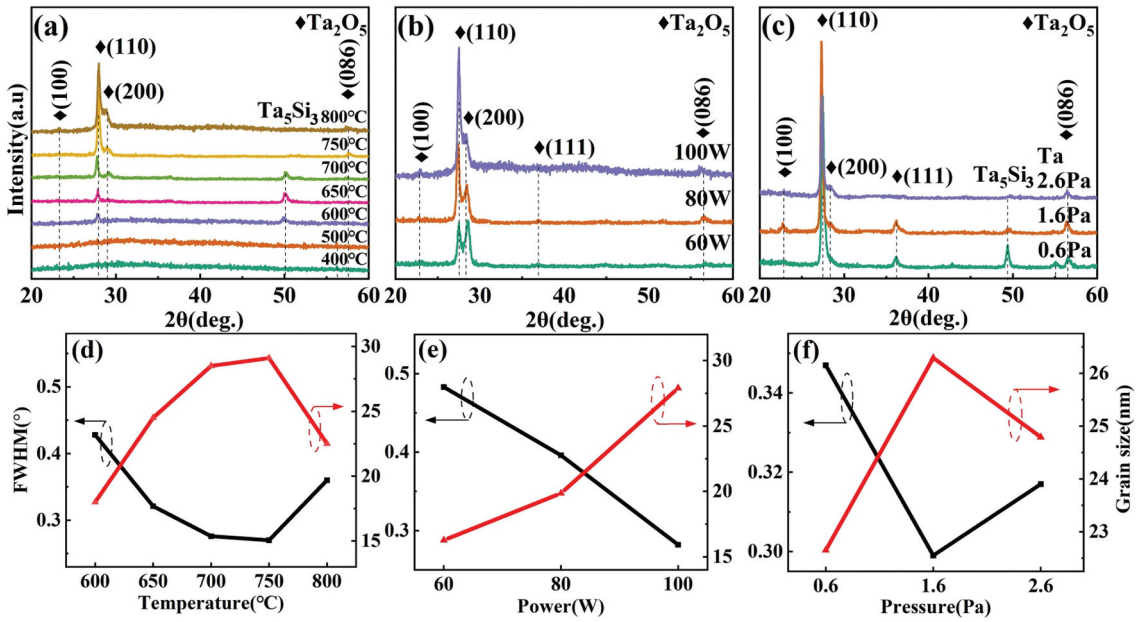


Figure 1. (a)~(c) XRD patterns of the TaO_x films prepared at different process conditions. (d)~(f) FWHM and grain size of the peak of (110) surface of these TaO_x films.

the substrate surface. As shown in Figure 1(b), XRD patterns of TaO_x films deposited at different RF powers (60, 80, and 100 W) were compared, with a temperature of 750°C and pressure of 1.0 Pa. With the increase of RF reactive sputtering power, the diffraction peak of (110) orientation gradually increases while the FWHM decreases. The increasing strength of the peak and the narrowing of FWHM indicate an enhanced crystallinity of the Ta_2O_5 film [26]. In addition, the evolution of grain size with deposition power is also shown in Figure 1(e). The grain size was calculated by Scherrer formula $D = 0.9\lambda/(W\cos\theta)$ [27]. The obtained results reveal that higher power promotes the grain growth of Ta_2O_5 films. Actually, higher power can provide greater kinetic energy for the migration, nucleation and growth of sputtered atoms/molecules on the substrate surface. Moreover, higher power tends to facilitate the sufficiency of reactive sputtering, which is also beneficial to the crystallization of the sputtered films. In addition, the influence of deposition pressure was also investigated. As shown in Figure 1(c,f), under the optimal conditions of temperature (750°C) and power (100 W), films deposited at 1.6 Pa exhibits the best crystallization characteristics.

Based on the above results, we started our main work on depositing TaO_x films at a temperature of 750°C, a growth pressure of 1.6 Pa and a power of 100 W, with a focus on investigating the effect of oxygen partial pressure. The reactive sputtering atmosphere is a critical processing parameter for depositing TaO_x film utilizing a Ta target. The oxygen partial pressure, or oxygen concentration in the atmosphere, directly determines the oxygen ion content in the oxide film, i. e. x value in TaO_x. Many attempts have been made to fabricate high-quality TaO_x-based films/devices and improve their properties. However, most existing research has focused on multilayer heterostructures, such as $Ta_2O_5/TaO_x/TiO_2$ [28], TaO_x/HfO_2 [29,30], and TaO_x/AlN [31], in consideration of their extrinsic contributions. Moreover, in many sputtering experiments, Ta_2O_5 ceramic targets with standard stoichiometric ratios are used, resulting in ineffective control over the composition of the prepared TaO_x film layer [20–23]. Even though Ta metal target is used in reactive sputtering, very few literatures put their attention to the intrinsic contribution of the TaO_x film itself. In other words, the previous investigations are basically based on single component TaO_x films and the x value of TaO_x is less than 2.5 [23,32,33]. In this work, TaO_x films were deposited on Si and ITO bottom electrode buffered Si substrates at different oxygen partial pressures (O_2 concentration in mixed O_2/A_r), specifically 10%, 15%, 20%, 25% and 30%.

SEM images of the TaO_x films deposited in different atmosphere are presented in Figure 2(a). Grain growth can be clearly seen on the natural surface of the high-density deposited film, and no cracks are present in the sample. Notwithstanding, TaO_x films prepared at relatively low oxygen partial pressure ratios, especially 10% – 15%, display an inhomogeneous surface morphology with obvious undulation. As the oxygen

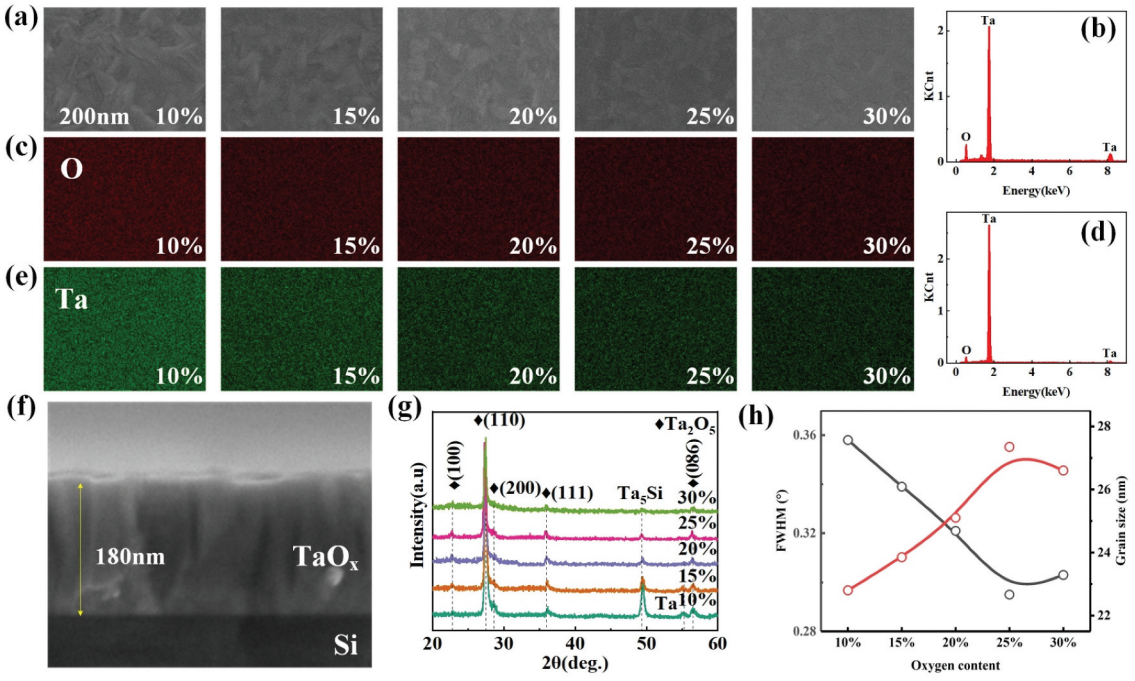


Figure 2. (a) SEM surface morphology and (f) cross-sectional topography of the TaO_x films prepared at different oxygen partial pressure. EDS mapping of (c) O and (e) Ta of these TaO_x films, (b) and (d) is the corresponding EDS composition diagram of the TaO_x films prepared at 25% and 30% O_2, respectively. (g) XRD patterns, (h) FWHM and grain size of the peak of (110) surface of the TaO_x films.

partial pressure ratio increases, the uneven structure gradually decreases. TaO_x films prepared at high oxygen partial pressure ratios, such as 25% and 30%, possess a smooth, dense and uniform microstructure. It also can be seen from the SEM cross-sectional image of the TaO_x film deposited at 25% oxygen partial pressure (Figure 2(f)). This indicated that high oxygen partial pressure is conducive to the improvement of the density and uniformity of TaO_x films, which is consistent with XRD pattern results. As confirmed by the XRD analysis in Figure 2(g), TaO_x films prepared at 10% and 15% O_2 ratio exhibit obvious Ta_5Si_3 impurity phase. The peak intensity decreases with the increase of O_2 ratio because increased oxygen content enables the oxidation reaction to involve more Ta and from the target film on the substrate. As shown in Figure 2(h), the grain size increases with increasing O_2 ratio, reaching a maximum at the 25% ratio. However, as the oxygen partial pressure gradually increases to 30%, the grain size decreases, and FWHM has an opposite relationship with the grain size. Therefore, in agreement with the conclusion of the paper, the TaO_x thin film grown at a 25% oxygen partial pressure has the largest grain size and the best crystallinity.

The spatial distribution of O and Ta elements in these films on Si substrate was also analyzed by the EDS mapping technique and the results are shown in Figure 2(c,e), respectively. The EDS spectra were also shown in Figure 2(b) (25% O_2 sample) and Figure 2(d) (30% O_2 sample) to confirm the presence of Ta and O elements in the prepared samples. The color mapping confirms the homogenous distribution of Ta and O elements in all TaO_x samples. Additionally, in the 10% O_2 sample shows the highest localized distribution of Ta elements, which is attributed to the production of more ionized Ar^+ at high argon partial pressure to bombard the Ta target, resulting in more Ta deposition on the substrate. At the same time, oxygen participation in the oxidation of densely distributed Ta deposited on the substrate surface at high temperature (750°C here) results in the highest density of oxygen distribution the 10% O_2 sample (the 1st picture in Figure 2(c)). With the increase of oxygen partial pressure, the distribution density of Ta elements decreases gradually. Increased O_2 pressure is associated with decreased Ar pressure, leading to an increase of neutral O atoms in the plasma. These neutral O atoms collide with the sputtering species, consuming the energy of the sputtering species. Consequently, these species (i.e. Ta) may not have enough energy to reach the substrate surface, resulting in less distribution. In summary, the oxygen content in the dielectric film plays a key role in the resistance switching (RS) process, which will strongly affect the RS characteristics and mechanism of TaO_x thin film devices.

To investigate the RS properties of the above TaO_x films prepared under different oxygen partial pressures, an ITO layer was prepared on a Si substrate as a bottom electrode prior to TaO_x layer deposition. In Figure 3, the color mapping of element distribution of the TaO_x/ITO/Si structures are shown. EDS mappings in Figure 3(a,b) confirm that there is a homogenous distribution of O and Ta elements in the TaO_x/ITO heterostructures. Comparing the EDS mapping images in Figures 2 and 3, it can be found that the variation trend of the element distribution density in TaO_x film prepared on ITO is different from that on Si. This is primarily attributed to the influence of the oxygen distribution in ITO and the diffusion between ITO and TaO_x. Figure 4(a,e) reveals the surface SEM images of the TaO_x/ITO films, which manifests that all films display a uniform and dense surface. The film prepared at 10% O_2 ratio was basically composed of long-strip grains. When the oxygen partial pressure was increased to 15%, a small amount of ellipsoidal grains appeared, the film was mainly composed of both strip and ellipsoidal-like grains. With the further increase of the oxygen partial pressure, more spherical/ellipsoidal-like grains were observed. Figure 4(f) describes the typical SEM cross-sectional image of the TaO_x/ITO heterostructure, where the 80-nm-thick ITO electrode layer exhibits a dense structure without obvious holes or cavities at the interface, which can promote the crystal growth of TaO_x, as confirmed by the columnar structure of TaO_x.

Figure 5 shows the current – voltage (*I*–*V*) characteristic curves of Ta/TaO_x/ITO devices fabricated for different oxygen concentration (10%, 15%, 20%, 25%,

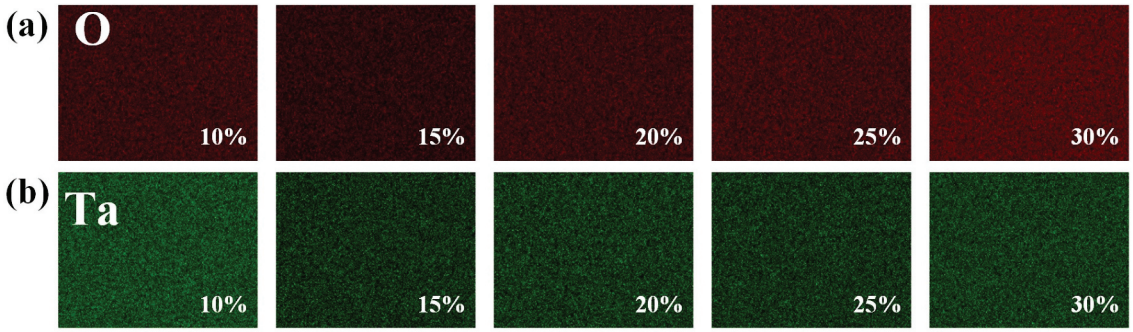


Figure 3. EDS mapping of (a) O and (b) Ta of the TaO_x/ITO films prepared at different oxygen partial pressure.

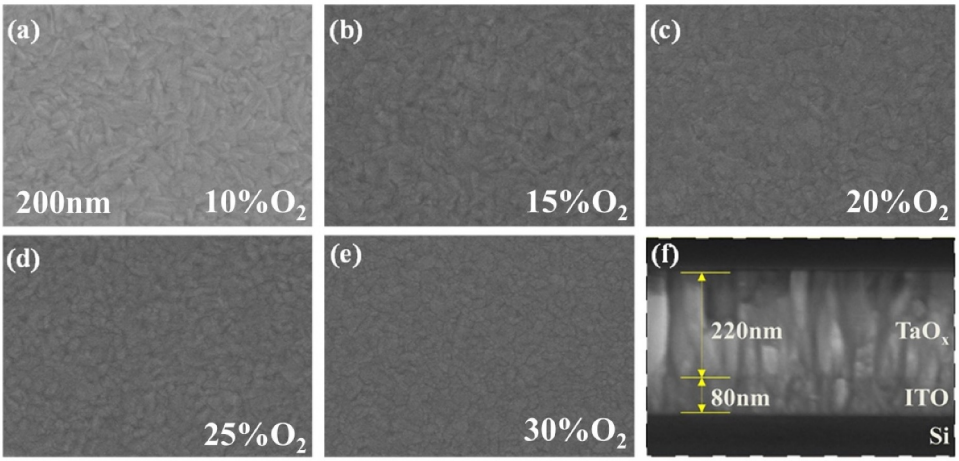


Figure 4. (a)~(e) SEM surface morphology and (f) cross-sectional topography of the TaO_x/ITO films prepared at different oxygen partial pressure.

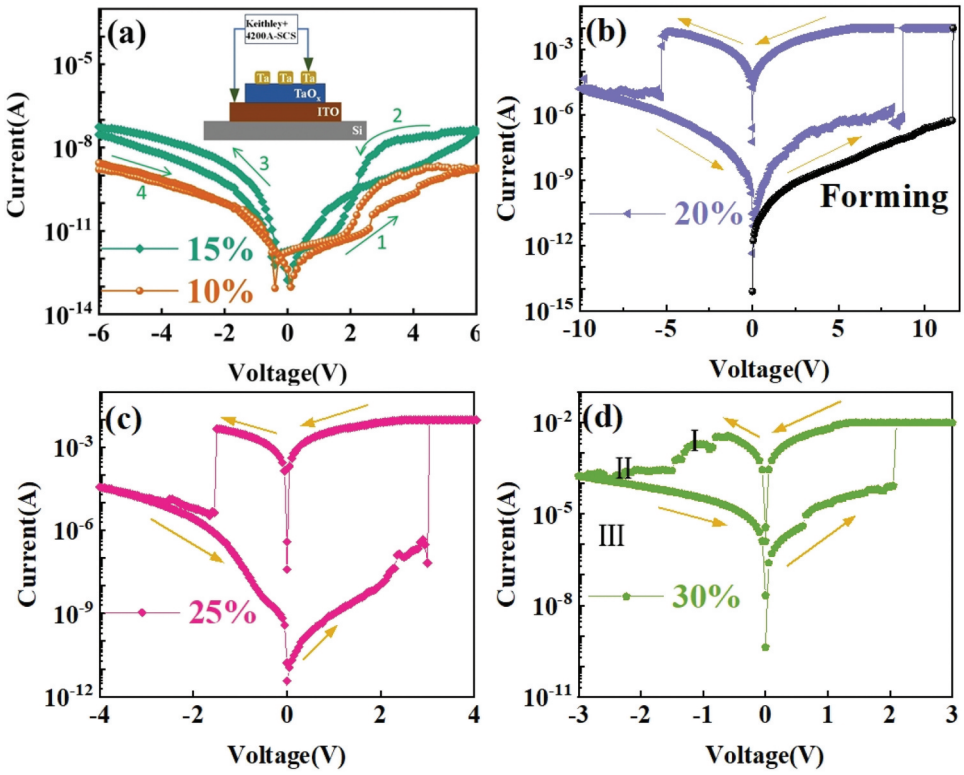


Figure 5. Typical I-V curves with different oxygen concentration of the Ta/TaO_x/ITO structures at room temperature, the arrows indicate the voltage sweeping direction. The Inset of (a) is a device model diagram of the Ta/TaO_x/ITO structures.

30%) of TaO_x layers. All *I-V* curves were measured through a tungsten-alloy probe in direct contact with the TE of the Ta/TaO_x/ITO device, while the BE was grounded. The corresponding schematic drawings of the sandwich structure are shown in the inset of Figure 5(a). It can be seen in Figure 5(a) that as the voltage sweeps from zero to a positive (negative) value, the corresponding current increases gradually and continuously. During the SET and RESET processes, the transition between high resistance state (HRS) and low resistance state (LRS) of the devices is gradual and continuous, indicating that the current is homogeneous throughout the device and the switching occurs in the entire TaO_x films fabricated at 10% and 15% oxygen concentration, which presents a typical homogeneous RS behavior [34]. However, when the oxygen partial pressure is increased (20%, 25%, and 30%), as shown in Figures 5(b,d), obvious bipolar characteristics are observed in the devices. The compliance current was set at 10 mA to prevent the permanent breakdown and destruction of the structures during the sweeping processes. The devices states are switched to LRS when the voltage is swept from 0 to the SET voltage (V_{set}), and then switched to HRS when the voltage is swept from 0 to the RESET voltage (V_{reset}). Thus, the RS behavior is achieved in Ta/TaO_x/ITO devices due to the

formation (LRS) and rupture (HRS) of the CFs in the oxide heterostructure. Furthermore, it becomes difficult for the device with 30% oxygen concentration to achieve the RESET process. As is depicted in Figure 5(d), a lot of time is wasted when the current decreases gradually from $V = -0.9$ to -2.8 V, and the RESET switching speed is slower than that of other oxygen concentration. This may be because the number of tantalum ions in the resistive layer is reduced, and the mobility of tantalum ions is also weakened [35]. On the basis of the above analysis, Ta/TaO_x/ITO devices with different oxygen concentration (20%, 25%, 30%) exhibit excellent RS characteristics, and the corresponding V_{set}, V_{reset} and forming voltage are shown in Figure 6(a) and summarized in Table 1. With the constant increase of oxygen concentration, the forming voltage, V_{set} and V_{reset} of the device decrease, and the positive SET voltage is always greater than the absolute value of the negative RESET voltage. This indicates that the oxygen content of TaO_x resistance layer has a great influence on the power consumption of the device, and appropriately increasing the oxygen content of TaO_x thin film can reduce the power consumption of the device. In addition, the HRS/LRS switching ratio and durability are two significant parameters for assessing the performance of TaO_x RS memory. As is depicted in Figure 6(b,d), the fabricated devices are tested 100 cycles without interruption between V_{set} and V_{reset} to verify the RS effect and durability at room temperature. As the oxygen concentration increases, the resistance of both HRS and LRS continues to decrease. This may be due to the O/Ta ratios of TaO_x with lower oxygen concentration being closer to that of Ta_2O_5, which has lower conductivity than other tantalum oxide, such as TaO and TaO_2 [36–38]. With an increase in the cycle number, the resistance remains almost constant, indicating favorable endurance characteristics.

To figure out the principle of electrical conduction, the *I-V* curves of Ta/TaO_x/ITO devices with different oxygen concentrations (25%, 30%) in log(|*I*|)-log(|*V*|) are plotted and fitted in Figure 7(a,c,e,g), As shown in Figure 7(a,e), the slopes of the double-logarithmic scale are approximately 1 in both HRS and LRS, indicating that the devices follow the ohmic conduction mechanism in region I and III. This is attributed to the

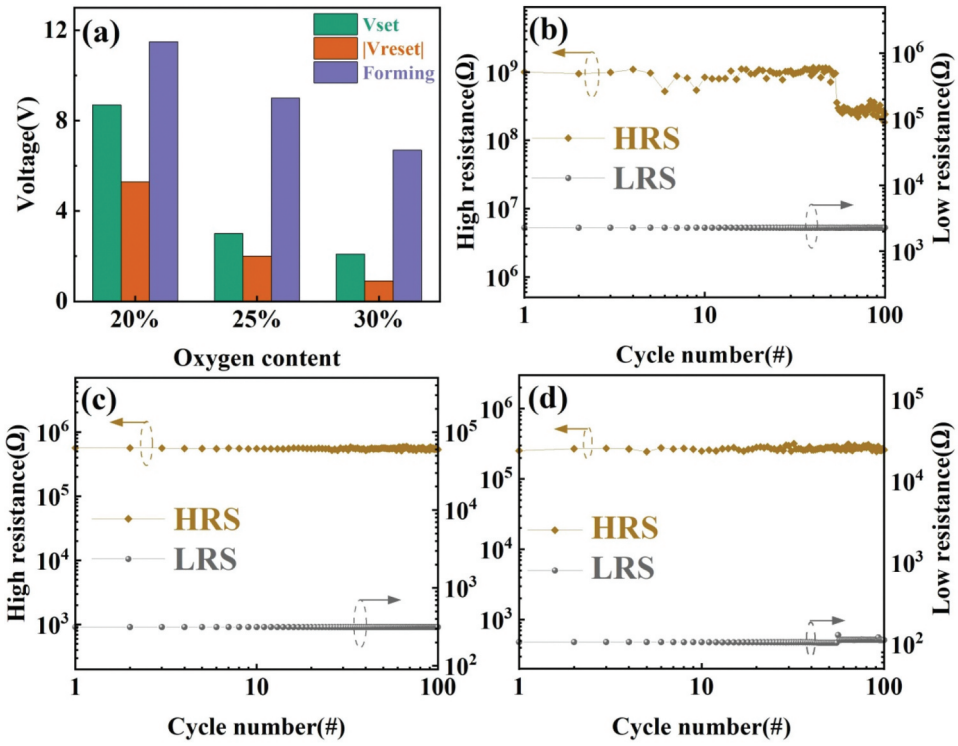


Figure 6. (a) Forming, setting and resetting voltages of the Ta/TaO_x/ITO devices with different oxygen partial pressure. (b)~(d) the corresponding resistance evolutions under high and low resistance status for different oxygen concentration (20%, 25%, 30%).

Table 1. Switching voltage, forming voltage, HRS/LRS ratio of the Ta/TaO_x/ITO structures with different oxygen concentration.

Structure	Ta/TaO_x/ITO		
Oxygen concentration	20%	25%	30%
V_{set}(V)	8.7	3.0	2.1
$\|V_{reset}\|$(V)	5.3	1.5	0.9
Forming voltage(V)	11.6	9.0	6.7
HRS/LRS ratio	10^5	1.8×10^3	2.4×10^3

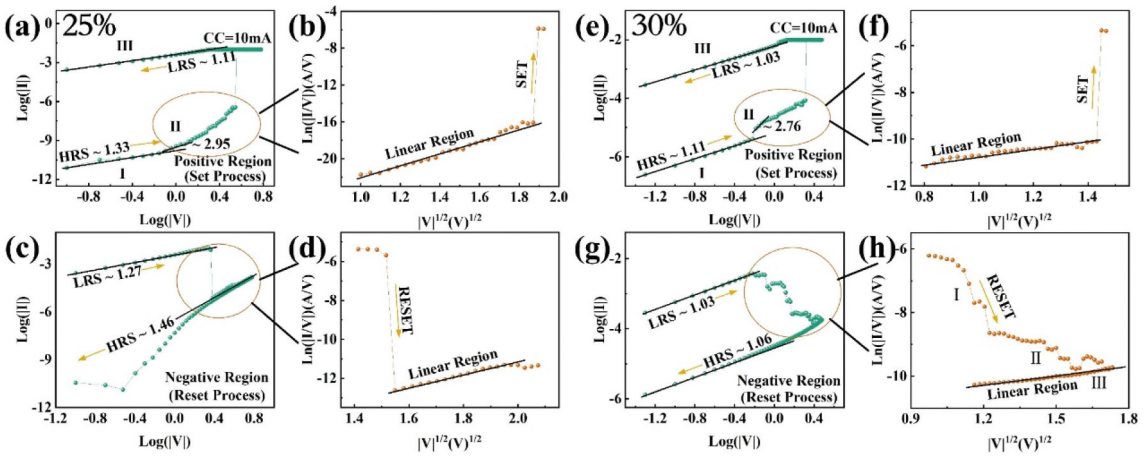


Figure 7. The fitted I – V curves in log(|I|)-log(|V|) and log(|I/V|)-$|V|^{1/2}$ scales of Ta/TaO_x/ITO structures in the positive-bias and negative-bias region with different oxygen concentration ((a)~(d) 25%, (e)~(h) 30%) at room temperature.

presence of more Ta^{5+} in the resistive layer under low voltage [27]. Moreover, the slopes of the log(|*I*|)-log(|*V*|) curve in the region II are 2.95 and 2.76 respectively. The device with 25% oxygen concentration exhibits a larger slope, potentially due to the significant impact of the number of Ta^{5+} ions on the resistivity of the TaO_x film [21]. However, the currents increase nonlinearly in the higher voltage regime. The nonlinear region plotted in a log versus log scale can be fitted further to create a linear curve in a ln(|*I*/*V*|) versus square-root of voltage plot depicted in Figure 7(b,d,f,h), which coincides with the Poole – Frenkel behavior [36]. As shown in Figure 7(b,d), a sharp increase (decrease) in current can be obtained at V_{set} (V_{reset}), and the device then switches to the LRS (HRS), which can be explained by the formation (rupture) of CFs. However, under the action of negative voltage, the RESET process of the structure prepared at 30% oxygen partial pressure becomes significantly slower, as shown in Figure 7(h). This can be attributed to the reduced number and mobility of the tantalum ions in the resistance layer. It is known that TaO_x films have abundant oxygen vacancies that play a significant role in conduction by trapping. To solve this problem, we refined the control of process parameters, based on the tantalum metal target and excessive oxygen content, combined with temperature control, to allow sputtered tantalum atoms/ions fully react with oxygen, generating a stable tantalum pentoxide phase (confirmed by XRD). This resulted in a significant reduction in the number of oxygen vacancy, and possibly even excess oxygen or free oxygen present in the dielectric film and its interface. When voltage is applied to the TE, the tantalum atoms in the electrode are oxidized into Ta^{5+} and migrate through the TaO_x film under the action of the electric field, where they are neutralized by electrons flowing from the cathode. Eventually, a conductive path connecting the top and bottom electrodes is formed when the Ta conductive filament becomes fully established.

The variation of resistance states is closely related to the change of CFs. By modeling the feasible RS mechanism (Figure 8), it is concluded that the formation and rupture of Ta conductive filaments, which

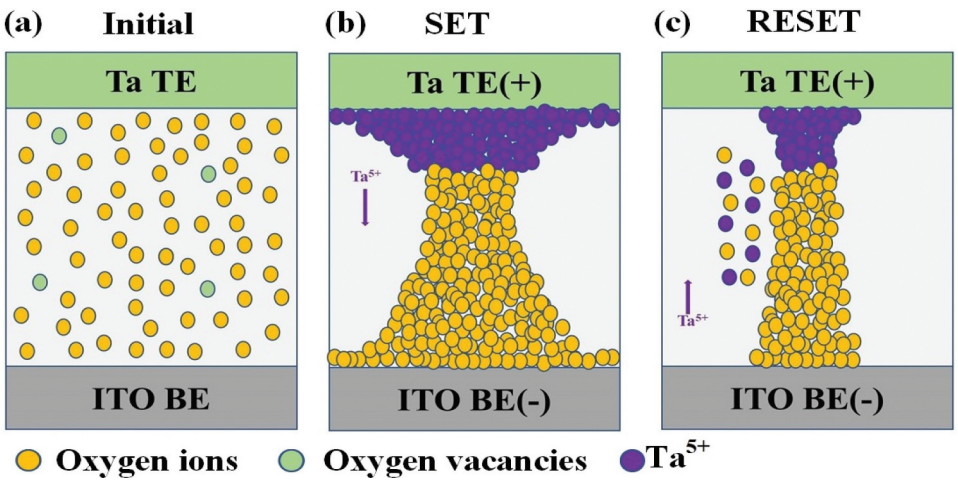


Figure 8. The schematic illustration for conductive filament mechanism. (a), (b) and (c) show the initial state, SET process and RESET process of the Ta/TaO_x/ITO structures, respectively.

results from electrochemical redox reactions and Ta^{5+} migration at the top electrode, can provide a better understanding of the conduction mechanism in our devices. The previous XRD results revealed that Ta is completely oxidized during the reactive sputtering process, and the grown thin film is a standard crystal film of Ta_2O_5 without any other tantalum oxide phases. Additionally, EDS results indicated that the Ta_2O_5 thin film is oxygen-rich. That is to say, there are very few oxygen vacancy defects in the prepared Ta_2O_5 film, and the formation/rupture of a conductive path cannot be attributed to the electro-migration of oxygen-related defects in metal-oxide insulating matrix. As shown in Fig .8(a), a large number of free oxygen ions exist in the TaO_x layer in the initial state without applied voltage. The electrochemical redox reaction of Ta atoms and the migration ability of the oxygen ions are weak when the applied voltage is low. As the positive voltage applied to the electrochemically active electrode Ta gradually increases to the V_{set}, Ta in the top electrode is reduced to Ta^{5+} and continuously migrates to the cathode through the TaO_x film, while negatively charged oxygen ions migrate to the TE interface, as shown in Fig .8(b). The current increases sharply when the applied voltage reaches V_{set}, indicating the formation of CFs formed by Ta ions and the device switches from HRS to LRS. When a sufficiently high negative bias is applied to the top electrode and the RESET process is cut, oxygen ions gradually are migrate back to the cathode and Ta conducting filaments are dissolved electrochemically, driving the device returning to the HRS (Fig .8(c)). Moreover, the formation and rupture of CFs may occur randomly within the TaO_x layer. As mentioned above, an increase in oxygen concentration leads to a segmented RESET process, as confirmed in Fig .5(d) and 7(h). It can be seen that when the oxygen partial pressure reaches 30%, the device resistance gradually changes during the RESET process, and the reset time increases obviously. However, excessively high oxygen concentration will lead to more joule heating, which is not conducive to the migration of tantalum ions. Therefore, excessive oxygen concentration can be detrimental to the RESET process of the device. In conclusion, the RS behavior in $Ta/TaO_x/ITO$ devices is closely related to Ta^{5+} migration and electrochemical redox reactions of Ta at the top electrode.

4. Conclusions

In summary, Ta_2O_5 films were prepared on Si (100) and ITO/Si (100) substrates by radio frequency reactive magnetron sputtering method. The sputtering process parameters, including deposition temperature, sputtering power, and gas pressure, were found to have a great influence on the crystallization characteristics. Based on the optimal process conditions (750°C, 100 W, and 1.6 Pa), high-quality TaO_x films were sputtered on ITO bottom electrode buffered Si substrate from a Ta target under different oxygen partial pressure ($O_2/(O_2+Ar)$: 10%, 15%, 20%, 25%, and 30%) by using RF-RMS. The oxygen content in the TaO_x layer was identified to have a significant impact on the RS characteristics and conduction mechanism of $Ta/TaO_x/ITO$ memory devices. The best crystallinity and RS characteristic were achieved in the device when the dielectric layer was prepared at 25% O_2. Research has shown that the O^{2-}/O concentration plays a leading role in improving performance. The component/microstructure optimized $Ta/TaO_x/ITO$ structure shows its great potential for RRAM application, as it exhibits excellent RS behavior. TaO_x films with 25% oxygen concentration exhibit the best crystallinity and, therefore, the best RS performance for their devices, including high durability ($>10^2$), high switching ratio (1.8×10^3), and fast switching in the RESET process.

Acknowledgments

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Disclosure statement

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